MINI PROJECT REPORT

Employee Attrition Prediction

Pattern Recognition & Principles (PRP)

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TOOLS UTILIZED IN PIPELINE

Python · Weka · Google Sheets · Flourish · Napkin AI

April 6, 2026

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1 Executive Summary

Project Overview

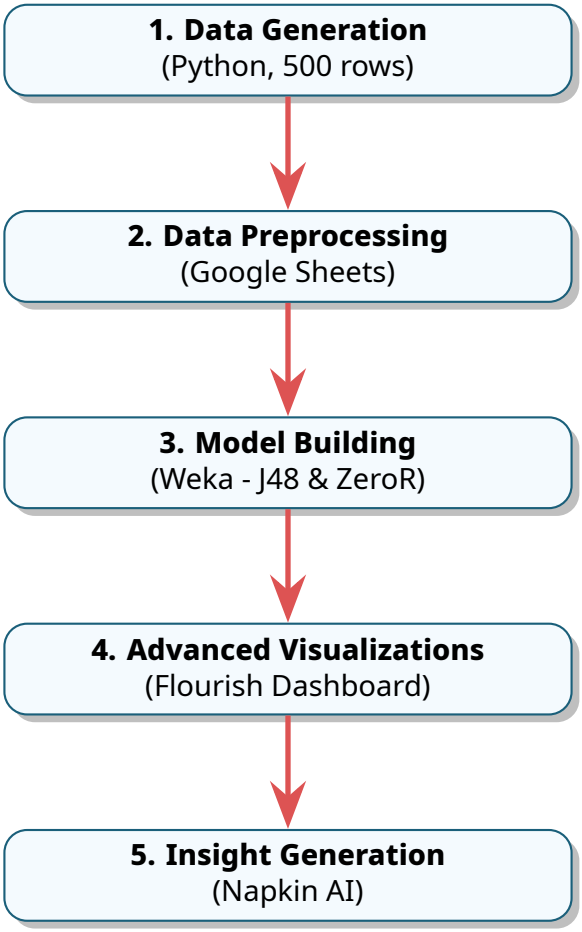
TechNova Solutions is currently facing a critical business challenge: a high rate of unexpected employee attrition. Finding and training replacements costs approximately 30-50% of an employee's salary, causing severe project delays and HR management issues. This project aims to build an end-to-end Machine Learning pipeline to predict **"Which employees are likely to leave?"** based on a meticulously generated dataset of 500 employees. The comprehensive workflow integrates data synthesis, rigorous cleaning mechanisms in Google Sheets, exploratory data analysis and modeling in Weka, advanced visual storytelling via Flourish, and AI-driven business insights via Napkin AI.

Important Project Links

- **Google Sheets (Raw & Cleaned Data):**
  Click Here to View Dataset Worksheet
- **Final Presentation Link:**
  Click Here to View Project Presentation

2 Project Workflow Methodology

The methodology of this project follows a structured data science lifecycle. We begin by synthesizing raw data, pass it through rigorous cleaning, deploy predictive classification models, and conclude by extracting visual and AI-driven insights for executive decision-making.



3 Data Preparation & Preprocessing

The Importance of Data Integrity

The foundation of any robust Machine Learning model lies in the quality of its data. The initial raw dataset was deliberately injected with inconsistencies to simulate real-world HR data challenges. Extensive data preprocessing was conducted to clean, format, and structure the data properly.

3.1 Raw Data Review

The initial raw dataset contained several formatting errors. For instance, there were duplicate entries and inconsistent capitalization across categorical variables (e.g., typing 'male ' instead of 'Male', or 'it' instead of 'IT').

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	EmployeeID	Age	Gender	Department	JobRole	MonthlyIncome	YearsAtCompany	YearsSinceLastPromotion	JobSatisfaction	WorkLifeBalance	EducationLevel	TrainingHoursLastYear	OverTime	PerformanceRating	Attrition
2	1001	50	Female	IT	System Admin	56844	3	0	4	3	3	33	No	5	No
3	1002	36	Male	Finance	Financial Analyst	79754	2	4	2	2	3	72	No	1	Yes
4	1003	29	Male	IT	System Admin	92129	8	9	3	2	1	27	No	3	No
5	1004	42	Female	Finance	Accountant	55042	7	4	3	3	3	10	Yes	2	No
6	1005	40	Female	IT	Software Engineer	76665	0	3	1	4	2	77	No	1	No
7	1006	44	male	Sales	Account Manager	131953	5	5	3	1	1	38	Yes	3	Yes
8	1007	32	Male	R&D	Research Scientist	133533	8	9	2	3	2	46	No	4	No
9	1008	32	Male	HR	HR Manager	88781	6	3	3	3	5	18	Yes	5	No
10	1009	45	Male	Finance	Accountant	81262	3	6	1	4	4	14	No	2	No
11	1010	57	Male	R&D	Research Scientist	61600	12	6	1	2	5	27	Yes	5	Yes
12	1011	45	Female	it	HR Manager	91898	16	1	2	4	4	23	Yes	4	No
13	1012	24	Female	HR	HR Manager	45448	1	2	2	3	5	71	Yes	3	Yes
14	1013	43	Female	IT	System Admin	41172	4	0	3	3	3	8	Yes	4	No
15	1014	23	Female	Sales	Account Manager	22657	11	7	3	2	2	46	Yes	4	Yes
16	1015	45	Male	Sales	Sales Executive	127855	20	8	1	2	4	32	Yes	5	Yes
17	1016	51	Male	R&D	Research Scientist	22906	7	3	2	2	1	61	No	2	Yes
18	1017	23	Male	Sales	Sales Executive	76413	5	5	2	4	2	18	No	4	No
19	1018	42	Female	Finance	Accountant	69028	7	8	2	2	1	66	Yes	5	Yes
20	1019	54	Female	Sales	Sales Executive	22287	17	5	3	3	2	73	Yes	2	No
21	1020	33	Male	R&D	Lab Engineer	48660	2	8	2	3	2	39	Yes	3	Yes

Figure 1: *Raw dataset containing duplicates and inconsistent case formatting*

3.2 Data Cleaning Steps (Google Sheets)

To ensure algorithmic accuracy, a systematic cleaning pipeline was executed directly in Google Sheets. The operations included deduplication, trimming trailing spaces, and structural normalization.

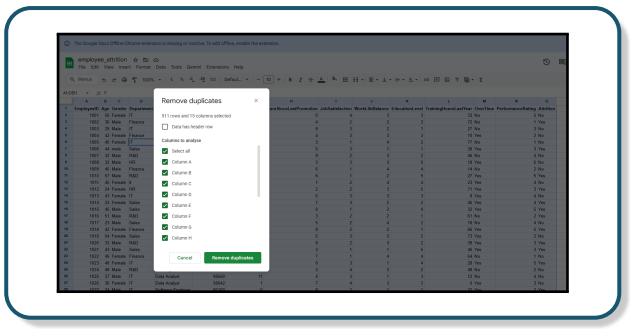


Figure 2: *Removing 10 intentional duplicates*

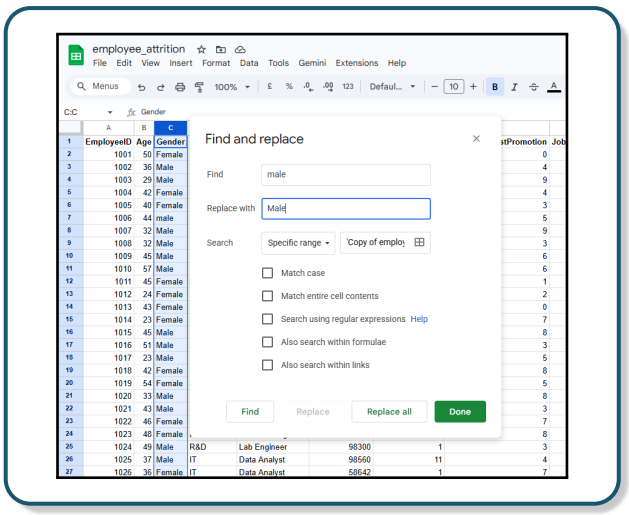


Figure 3: *Using Find & Replace to fix 'male '*

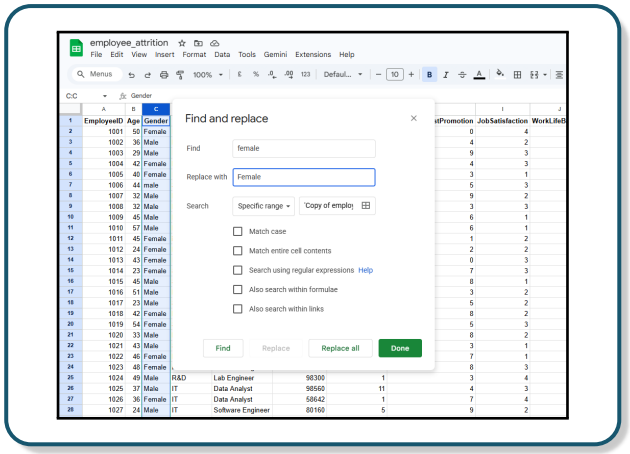


Figure 4: *Standardizing 'female' to 'Female'*

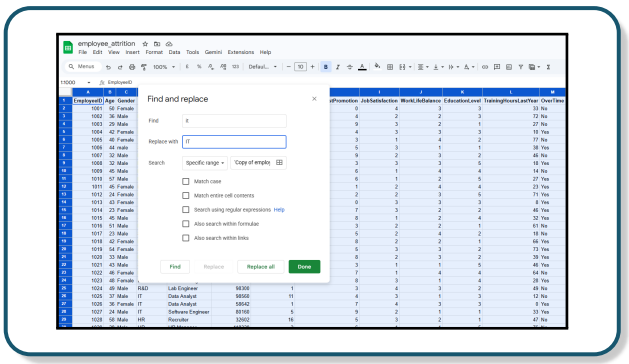


Figure 5: *Standardizing 'it' to uppercase 'IT'*

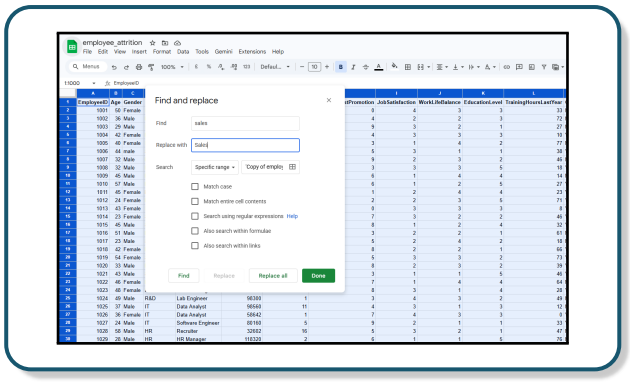


Figure 6: *Standardizing 'sales' to 'Sales'*

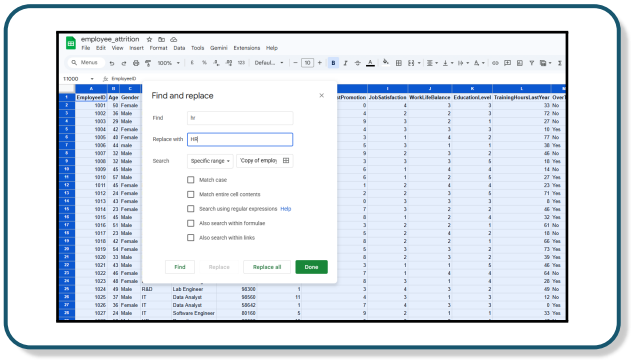


Figure 7: *Standardizing 'hr' to 'HR'*

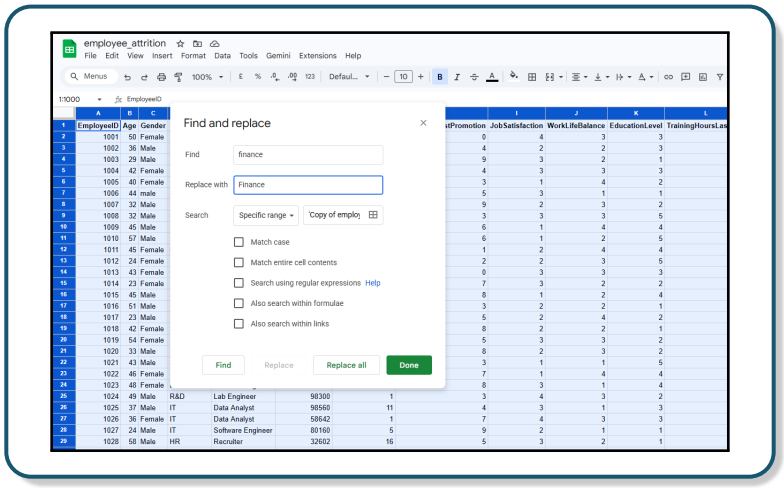


Figure 8: *Capitalizing 'finance' to 'Finance'*

3.3 Cleaned Data Result

After executing the cleanup pipeline, the resulting dataset was completely free of anomalies. We successfully generated exactly 500 perfectly structured records, primed for predictive modeling.

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
1	EmployeeID	Age	Gender	Department	JobRole	MonthlyIncome	YearsAtCompany	YearsSinceLastPromotion	JobSatisfaction	WorkLifeBalance	EducationLevel	TrainingHoursLastYear	OverTime	PerformanceRating	Attrition
2	1001	50	Female	IT	System Admin	56844	3	0	4	3	3	33	No	5	No
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21	1020	33	Male	R&D	Lab Engineer	48660	2	8	2	3	2	39	Yes	3	Yes

Figure 9: *Final 500-row cleaned dataset ready for Pattern Recognition*

4 Exploratory Data Analysis in Weka

Weka Environment Initialization

With the dataset fully sanitized, it was migrated into the Weka Machine Learning environment. The CSV file was first converted to the `.ARFF` format, which is the native and most optimized file structure for Weka's classification and visualization tools.

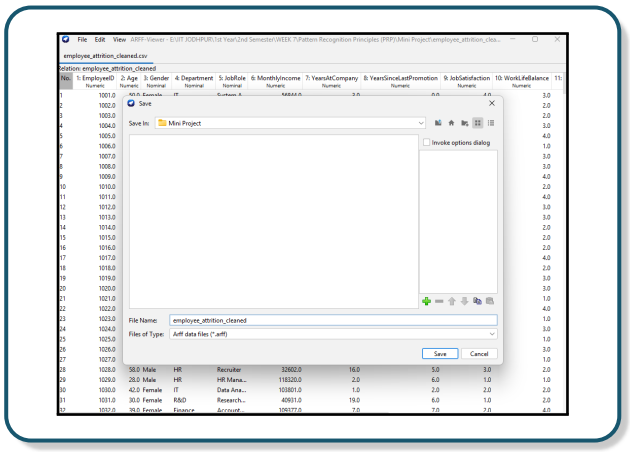

Figure 10: *Converting dataset to .ARFF format*

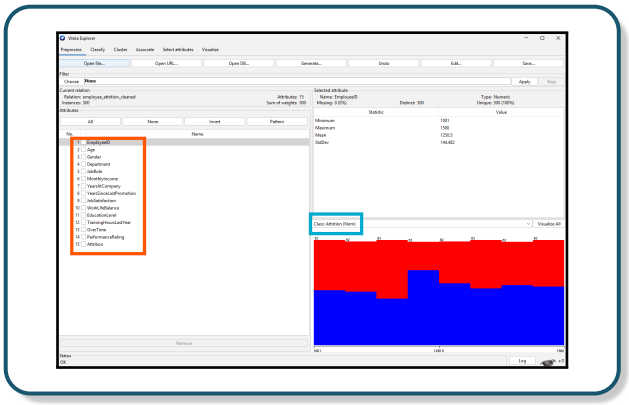

Figure 11: *Data loaded successfully in Weka Explorer*

4.1 Feature Visualizations

Before applying complex models, 2D scatter plots and matrices were plotted. This helps identify baseline correlations between the features (like Job Satisfaction and Overtime) and the target variable (Attrition).

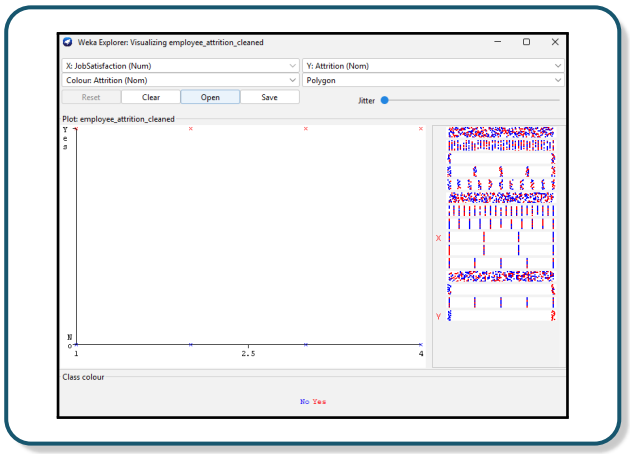

Figure 12: *Density of attrition at lower satisfaction levels*

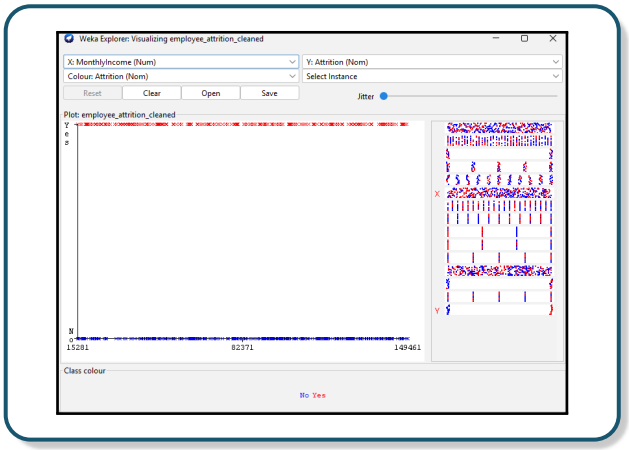

Figure 13: *Lower salaries corresponding to higher attrition*

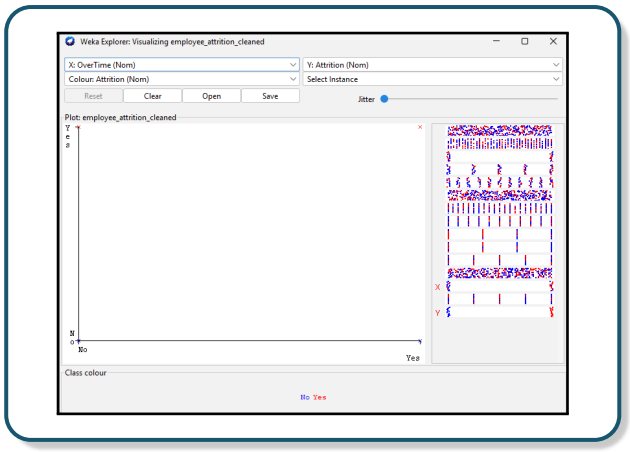

Figure 14: *OverTime="Yes" shows marked increase in attrition*

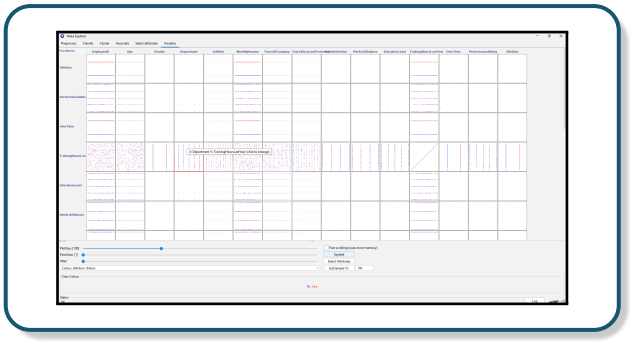

Figure 15: *Complete attribute matrix visualization*

5 Model Building & Evaluation (Weka)

To predict employee attrition, multiple classification algorithms were tested. The core objective was to identify a model that balances high accuracy with real-world generalization, while also empirically demonstrating the principles of **Overfitting** and **Underfitting**.

5.1 Baseline Model: J48 Decision Tree

A J48 decision tree classifier (an implementation of the C4.5 algorithm) was utilized as the baseline model. Decision trees are highly interpretable, making them perfect for HR datasets where explaining *why* an employee leaves is crucial.

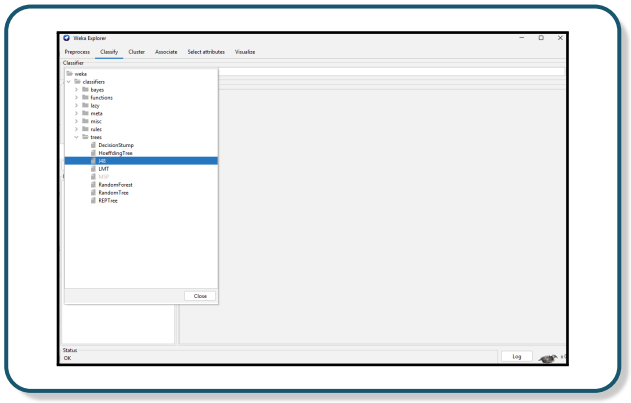

Figure 16: *Selecting J48 Tree classifier*

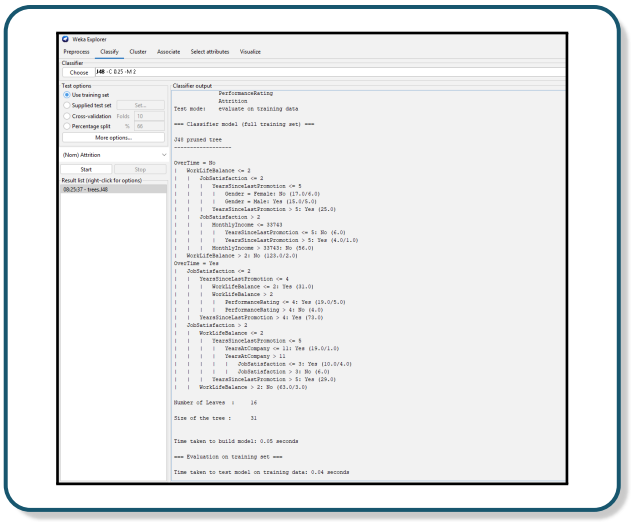

Figure 17: *J48 default configuration parameters*

J48 Baseline Results Interpretation

The J48 model performed exceptionally well on the standard parameters. It achieved an overall **accuracy of 94.6%**, correctly classifying 473 out of 500 instances. The confusion matrix validates that the model accurately predicted both "Yes" and "No" classes efficiently.

```
=== Summary ===

Correctly Classified Instances         473               94.6    %
Incorrectly Classified Instances        27                5.4    %
Kappa statistic                          0.8907
Mean absolute error                      0.0793
Root mean squared error                  0.1991
Relative absolute error                 16.0883 %
Root relative squared error             40.1113 %
Total Number of Instances              500

=== Detailed Accuracy By Class ===

                 TP Rate  FP Rate  Precision  Recall   F-Measure  MCC      ROC Area  PRC Area  Class
                 0.943    0.050    0.960      0.943    0.951      0.891    0.986     0.984     No
                 0.950    0.057    0.929      0.950    0.939      0.891    0.986     0.981     Yes
Weighted Avg.    0.946    0.053    0.946      0.946    0.946      0.891    0.986     0.983

=== Confusion Matrix ===

   a   b   <-- classified as
 264  16 |   a = No
  11 209 |   b = Yes
```

Figure 18: *Detailed Summary Output for Baseline J48 Classifier*

5.2 Overfitting Demonstration

Demonstrating Overfitting

Overfitting occurs when a model learns the training data *too* well, capturing noise instead of general patterns. To demonstrate this, the J48 parameters were deliberately altered (`minNumObj=1`, `unpruned=True`) to prevent the tree from pruning itself.

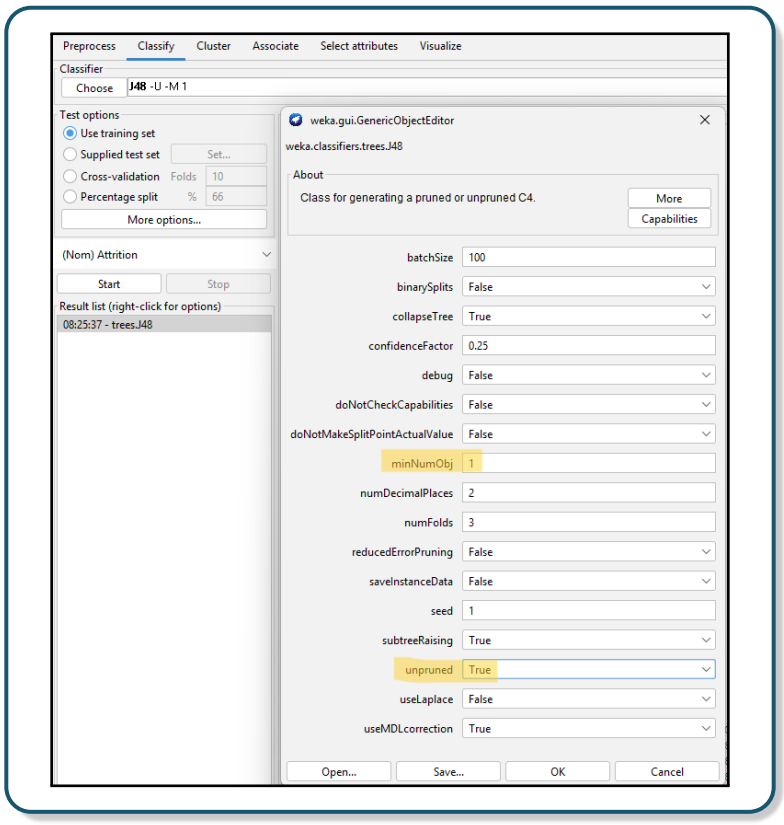

Figure 19: *Altering parameters to force model overfitting*

By forcing the decision tree to remain unpruned, it memorized every single data point. It achieved an unrealistic **100% accuracy on the training set**.

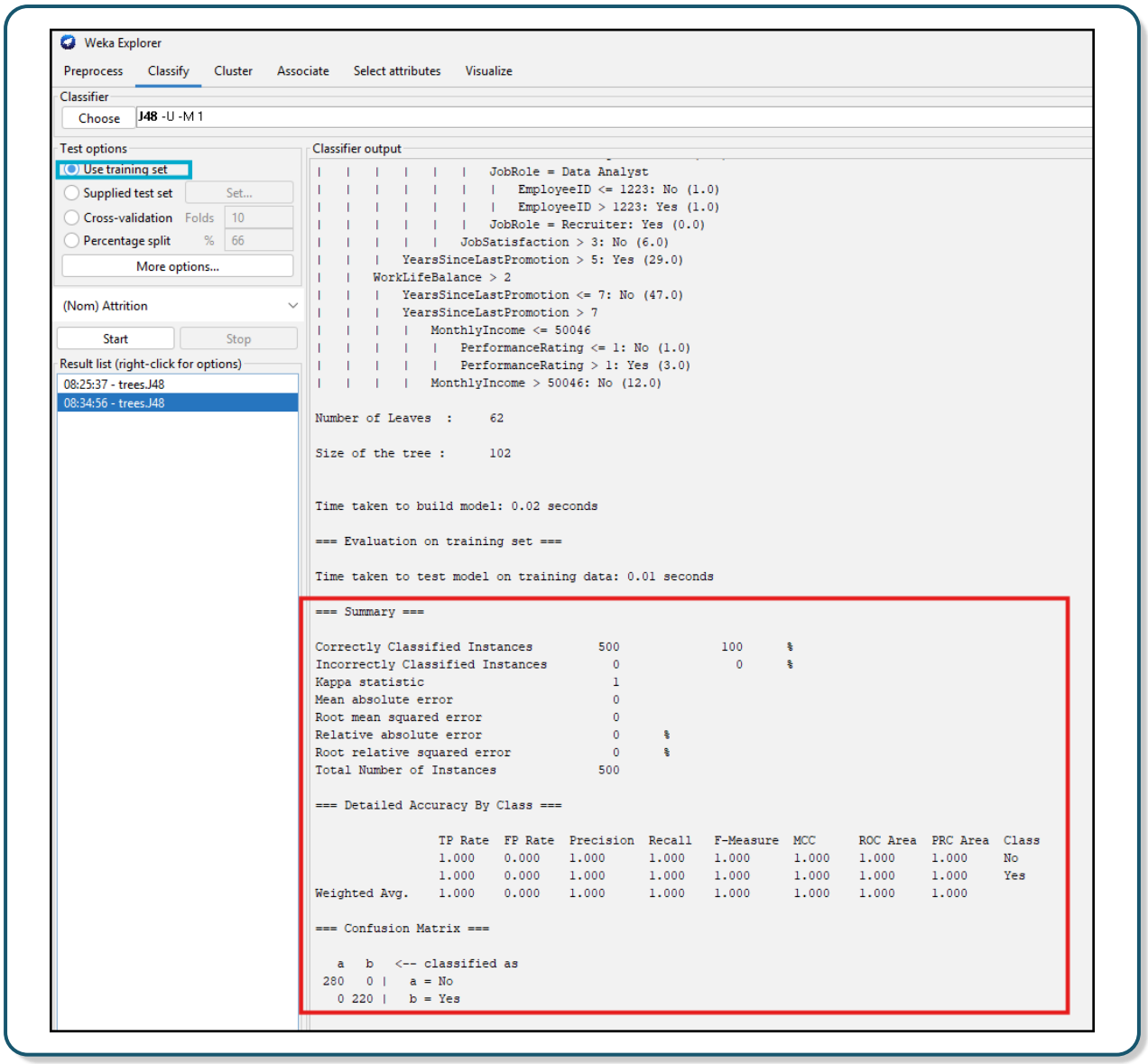

Figure 20: *Overfitting: Deceptive 100% Accuracy on Training Set*

However, when evaluating its true predictive power using Cross-Validation, the accuracy dropped sharply to **87%**. This empirically proves that an overly complex model fails to adapt to unseen data.

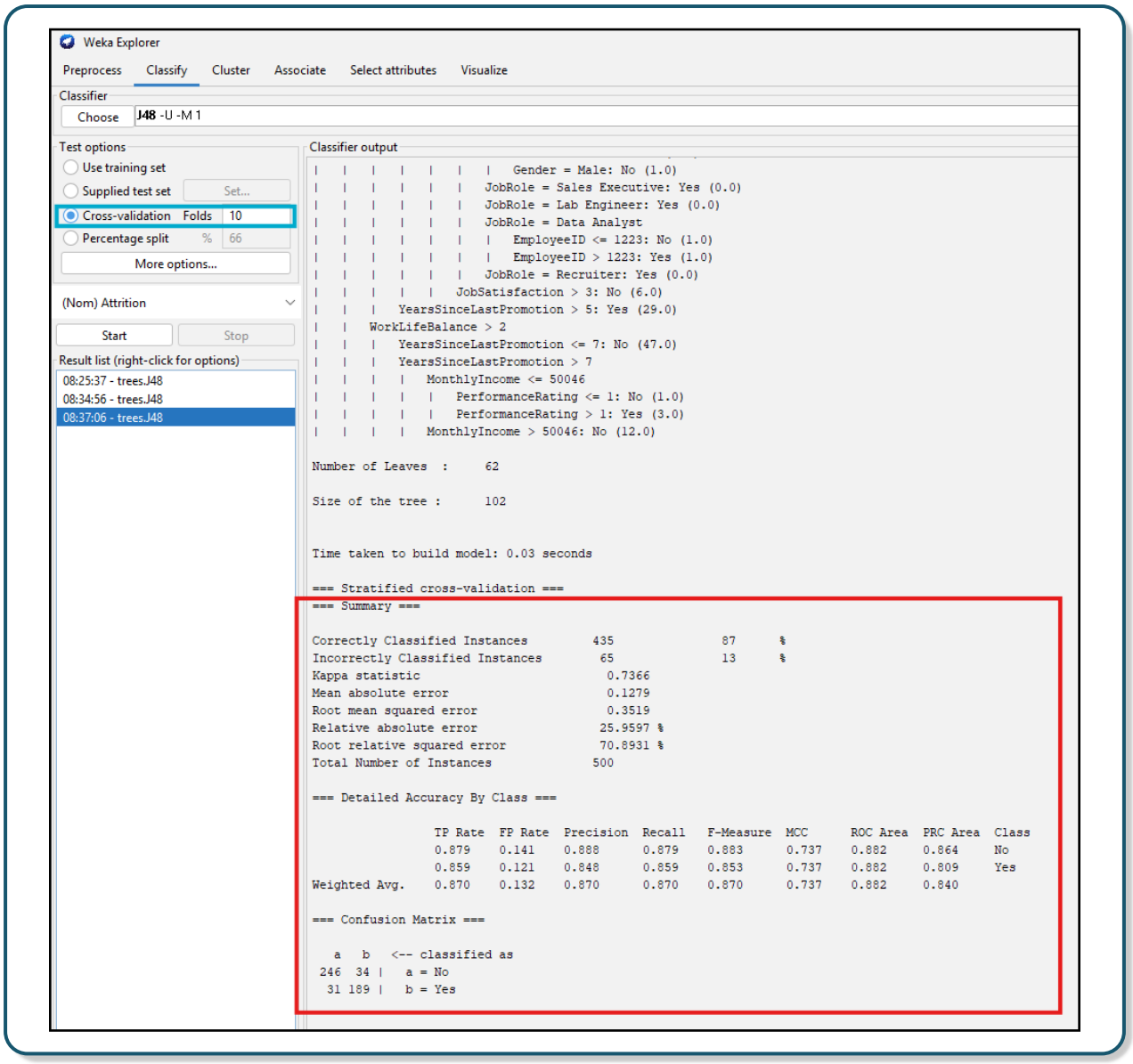

Figure 21: *Overfitting: Accuracy drops to 87% during Cross-Validation*

5.3 Underfitting Demonstration: ZeroR

Demonstrating Underfitting

Underfitting occurs when a model is too simple to capture the underlying patterns. The ZeroR classifier was used to demonstrate this; it ignores all input features and purely predicts the majority class.

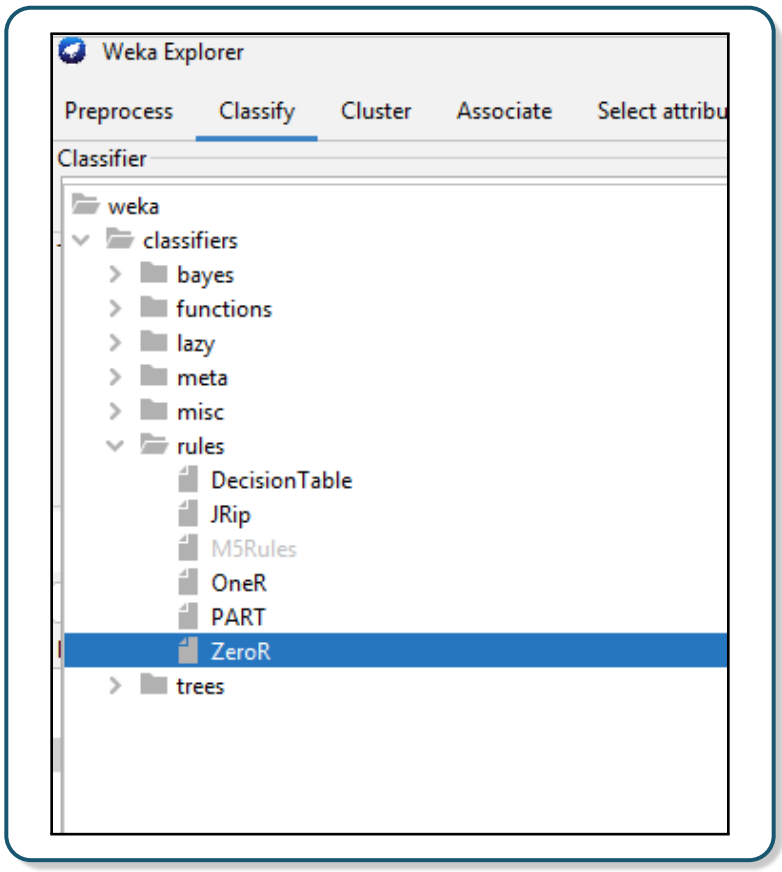

Figure 22: *Selecting the ZeroR rules-based classifier*

The model simply predicted "No" for all instances because "No" was the majority class in the dataset. It achieved a baseline accuracy of **56%**, completely failing to learn any underlying correlations from critical features.

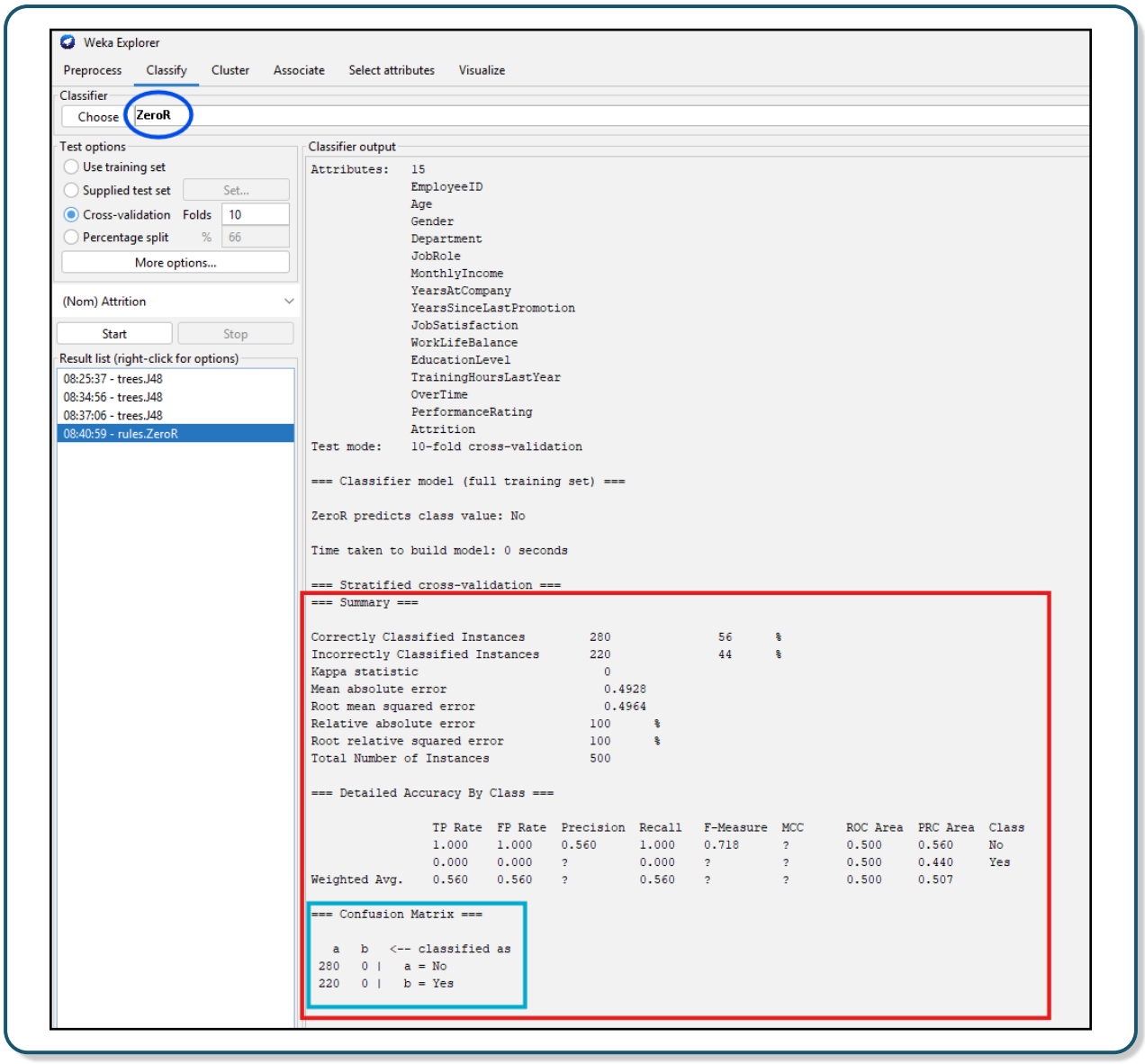

Figure 23: *ZeroR Output: Baseline majority-class prediction of 56%*

6 Advanced Visualizations (Flourish)

Executive Storytelling

While Weka provides excellent technical graphs, executive stakeholders require clean, modern dashboards. Pivot tables were utilized in Google Sheets to aggregate the data, which was subsequently imported into Flourish for premium visual storytelling.

6.1 Department-wise Attrition

The bar chart analysis clearly highlights that departments heavily reliant on targets or intensive workloads (such as HR, Sales, and IT) exhibit higher absolute attrition counts compared to the Finance department.

Department	COUNTA of Attrition
Finance	31
HR	48
IT	47
R&D	46
Sales	48
Grand Total	220

Figure 24: *Pivot table aggregation for Departments*

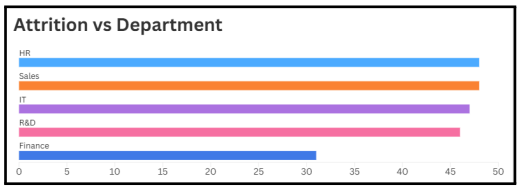

Figure 25: *Flourish Bar Chart visualization*

6.2 Job Satisfaction Analysis

The donut chart dramatically visualizes the psychological aspect of attrition. A vast majority of the employees leaving the company reported a job satisfaction level of 1 or 2, proving it is a core driver of attrition.

JobSatisfaction	COUNTA of Attrition
1	80
2	81
3	35
4	24
Grand Total	220

Figure 26: *Grouping by satisfaction levels*

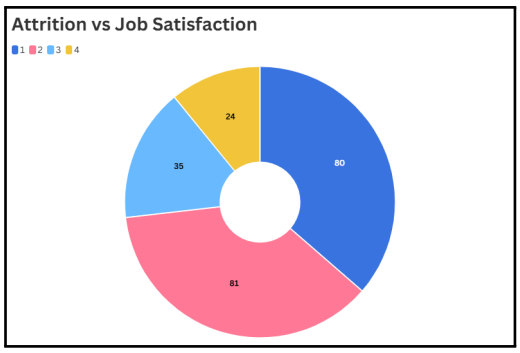

Figure 27: *Flourish Donut Chart visualization*

7 AI-Driven Insights (Napkin AI)

To bridge the gap between technical machine learning outputs and actionable business strategy, Napkin AI was prompted with the Weka model results. It successfully translated complex matrices into straightforward HR intelligence.

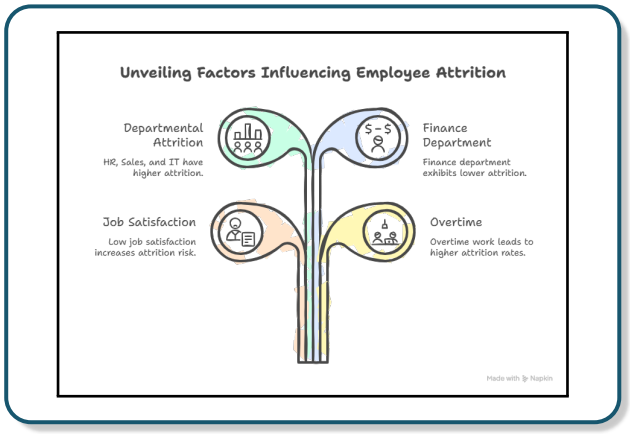

Figure 28: *Interpreting the initial Weka J48 results*

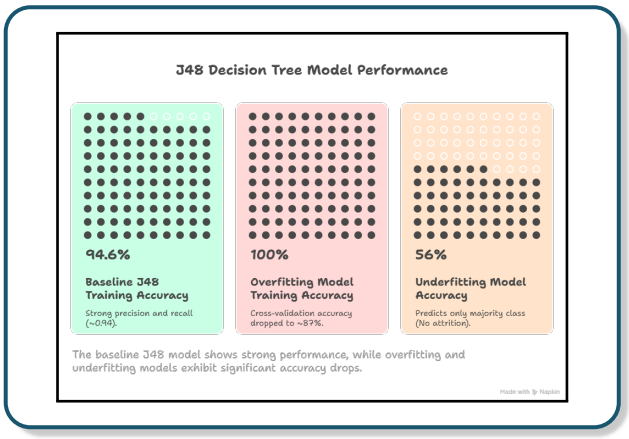

Figure 29: *Explaining Overfitting & Underfitting simply*

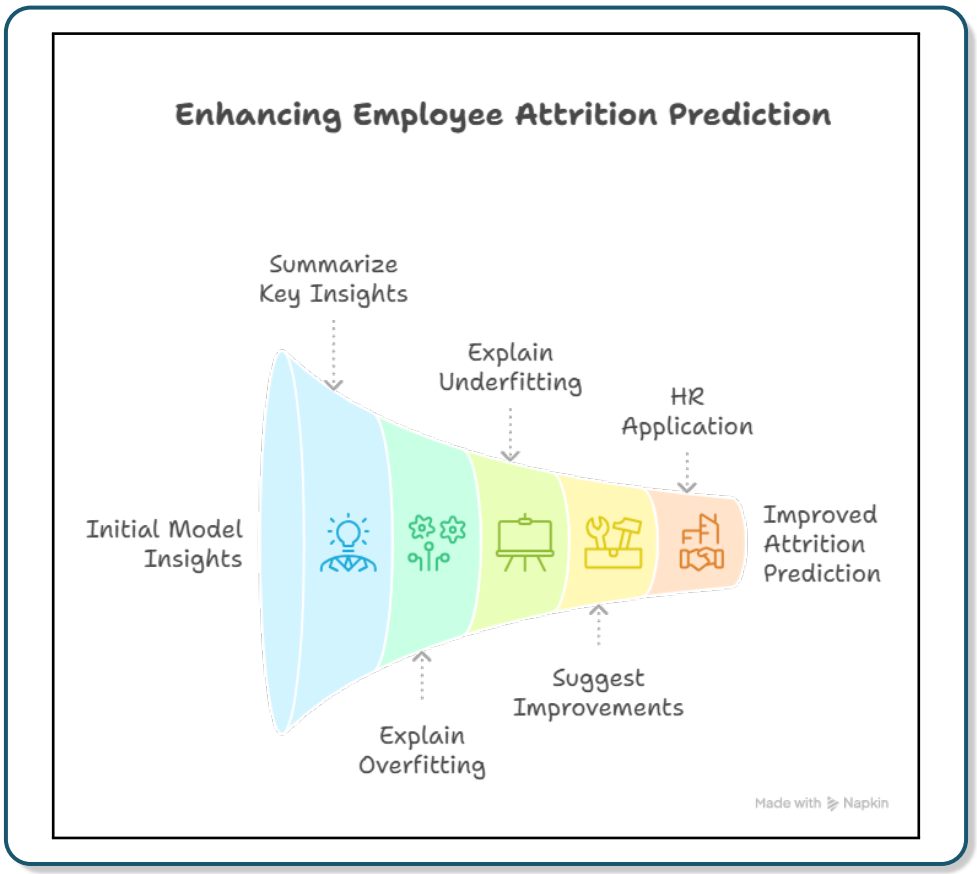

Figure 30: *Actionable HR recommendations generated by Napkin AI*

Key Insights

Based on the comprehensive data analysis, modeling, and AI interpretation, the following key insights were drawn:

1. **Core Drivers:** Low job satisfaction (Levels 1 & 2) and high overtime frequency are the strongest indicators of an employee's intent to leave.
2. **Financial Factors:** Employees earning lower monthly incomes are statistically more vulnerable to attrition.
3. **Departmental Stress:** The Sales and HR departments show concerning attrition volumes, hinting at systemic workload or management issues.
4. **Model Integrity:** The pruned J48 model is highly reliable (94.6% accuracy), proving that algorithmic prediction is a viable HR tool.

Final Conclusion

This mini-project successfully demonstrated the complete lifecycle of a Pattern Recognition and Machine Learning task. By tracking a simulated workforce's data, we established a robust pipeline capable of accurately predicting attrition.
The visual storytelling achieved through Flourish, combined with the technical rigor of Weka models (successfully identifying underfitting and overfitting), provides a clear, actionable framework. TechNova Solutions' HR department can immediately deploy these methodologies to proactively identify at-risk talent.

Reflection Questions Answered

Q1: What were the most important factors for attrition?
Based on the decision tree branches and visualizations, job satisfaction, overtime frequency, and monthly income were consistently the most significant predictive features.

Q2: How did the model react to parameter changes?
The model showed classic signs of overfitting when left unpruned (achieving 100% training accuracy but failing on cross-validation), and underfitting when using the overly simplistic ZeroR algorithm (56% baseline accuracy).

Q3: How can this model be improved further?
Adding more nuanced, real-world features (e.g., direct manager ratings, daily commute distance) and deploying advanced ensemble methods like Random Forest could further enhance real-world generalization.

Q4: How can HR practically utilize this model?
HR can integrate this predictive model into their quarterly reviews to silently flag "At-Risk" employees, allowing management to intervene proactively with timely promotions, pay raises, or workload reductions.

— END OF REPORT —